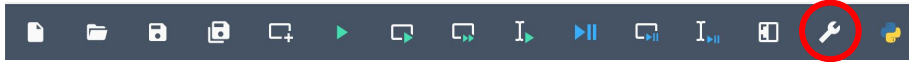


Plotting in Python

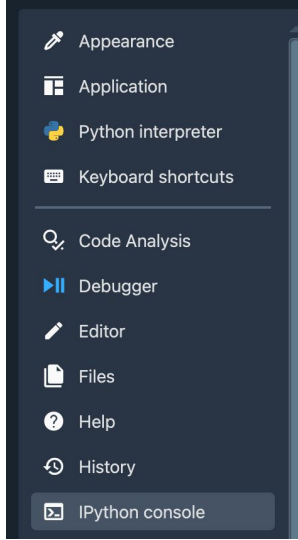
ReproRehab Pod2

If you want to see interactive plots in Spyder...

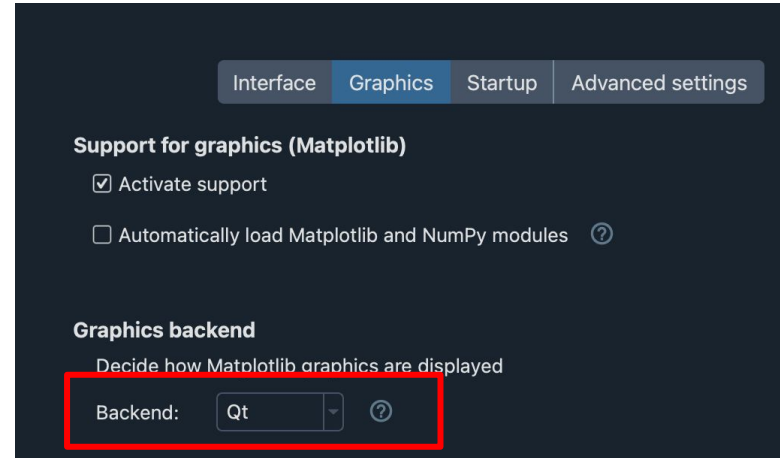
1. Click “Preferences”



2. Select “IPython console”

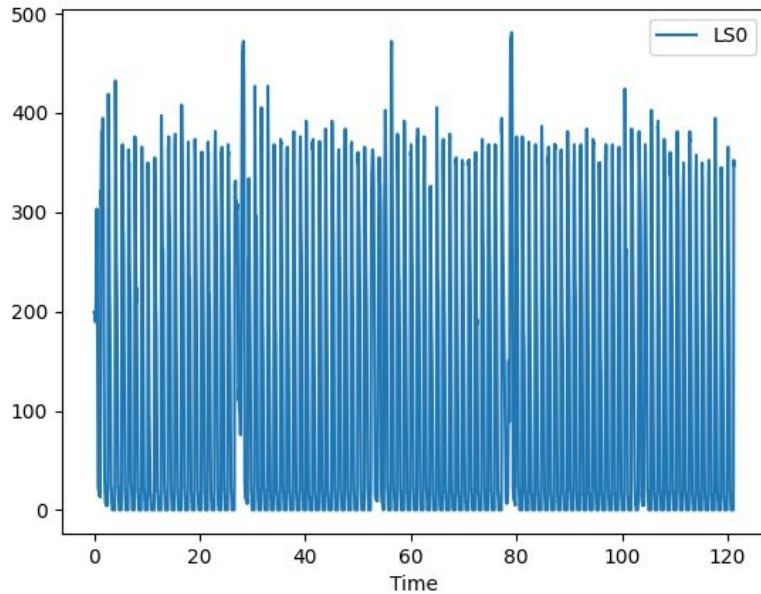


3. Select “Graphics” and change Graphics backend to “Qt”

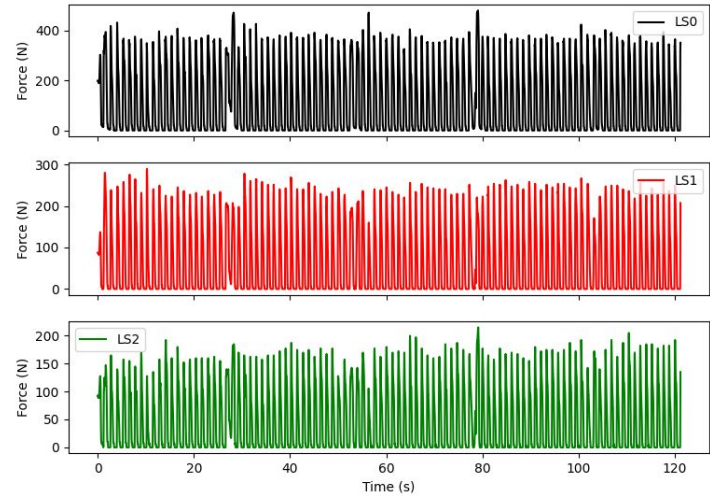


Recap - pandas can also help with drawing plots!

```
data.plot.line(x='Time', y='LS0')
```



```
data.plot.line(subplots=True,  
               x='Time', y=['LS0', 'LS1', 'LS2'],  
               xlabel='Time (s)',  
               ylabel='Force (N)',  
               color=['k', 'r', 'g'])
```



Use Matplotlib if you need more control

`matplotlib.pyplot.subplots()`

```
import matplotlib.pyplot as plt
```

```
fig, axes = plt.subplots(3, 2, figsize=(12, 6), sharex=False, sharey=True)
```

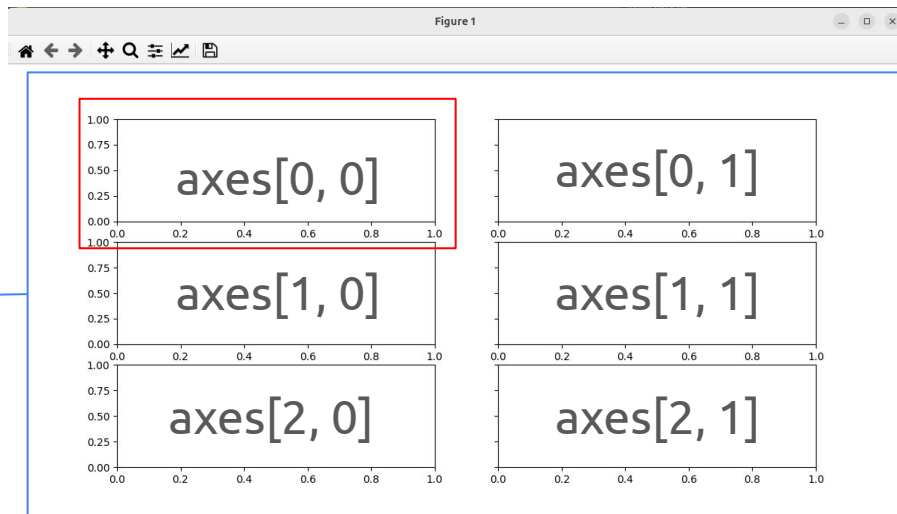
Width and height in inches

Number of columns

Number of rows

These are variables,
so any name you can
use (ex. 'Figure', or
'ax'.)

This is fig

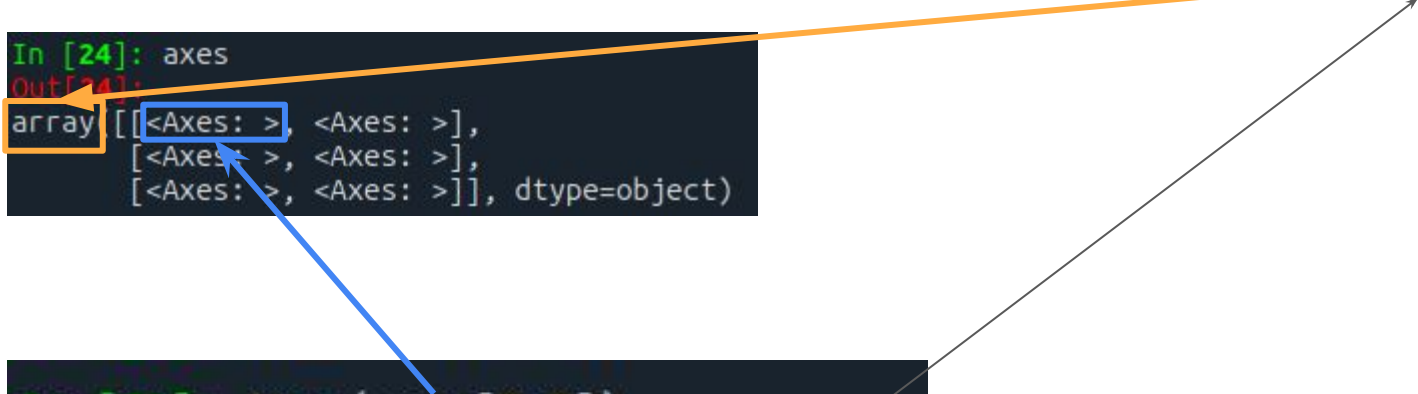


Little more detail (don't need to know this now!)

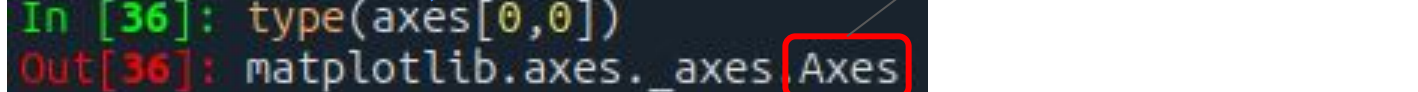
`fig, axes = plt.subplots(...)` <- This means the function returns two values.

If you check what `axes` is, you see that it's a **numpy ndarray** of **Axes**.

```
In [24]: axes
Out[24]:
array([[<Axes: >, <Axes: >],
       [<Axes: >, <Axes: >],
       [<Axes: >, <Axes: >]], dtype=object)
```

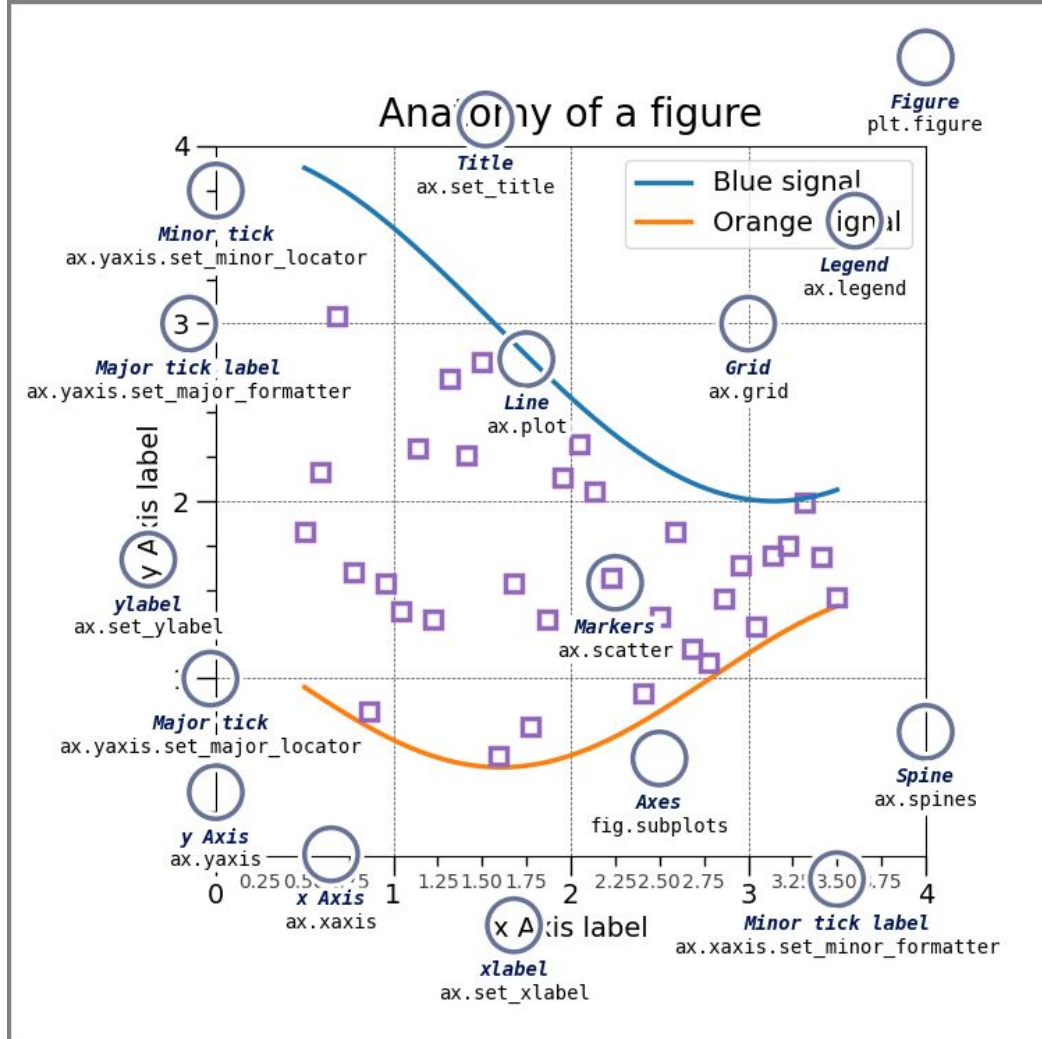


```
In [36]: type(axes[0,0])
Out[36]: matplotlib.axes._axes.Axes
```



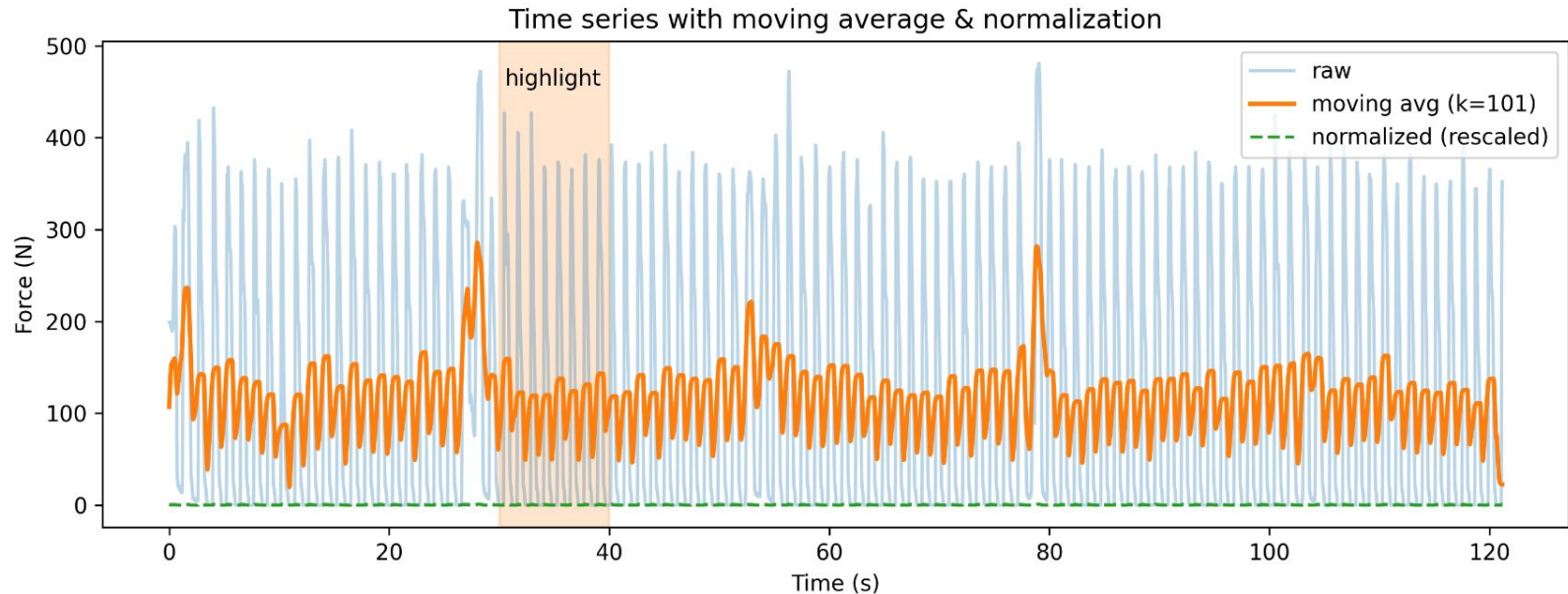
Anatomy of a figure

<https://matplotlib.org/stable/gallery/showcase/anatomy.html>



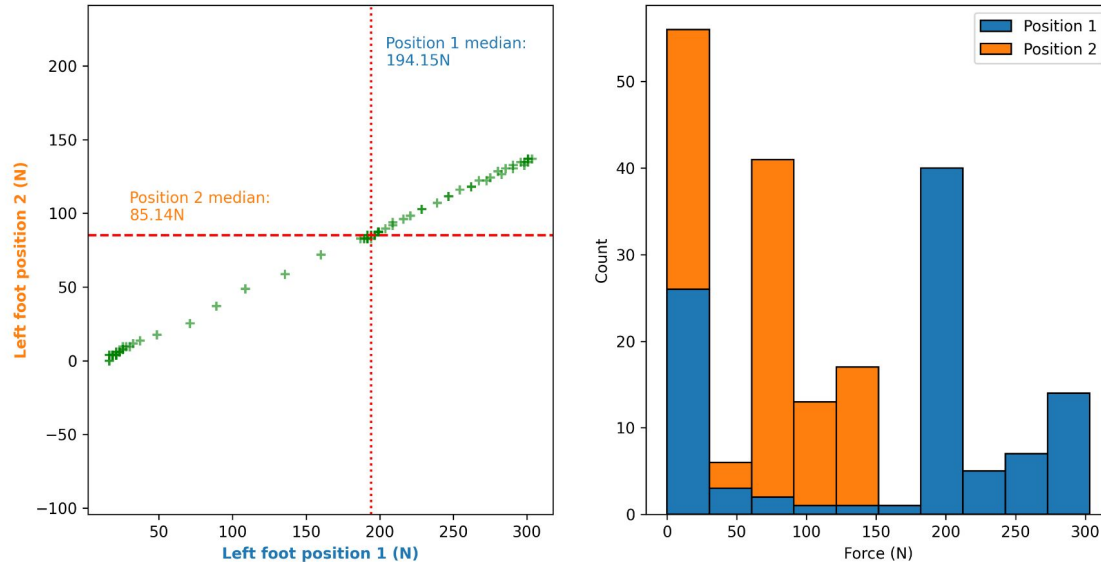
Assignment: make 3 figures

First figure: Time_series.png



Assignment: make 3 figures

Second figure:



Assignment: make 3 figures

Third figure: Custom_layout.png

